TEXORA · LECTURE SHEET

Methods and Standards for Evaluating Comfort and Protective Performance

How comfort and protection claims are proven: the three-level evaluation ladder — fabric instruments, thermal manikins, wearer trials — and the standard test methods behind the protective performance codes of Topic 3.

AM-4103 · CLOTHING COMFORT & SPECIAL APPAREL PRODUCTS · TOPIC 4 · LEVEL 4 TERM 1

1. The evaluation ladder

No single test proves a garment comfortable or protective. Evaluation climbs a ladder of increasing realism and cost, and a competent specification uses all three rungs:

LEVEL	METHOD	WHAT IT ADDS / WHAT IT MISSES
1 · Fabric	The Topic 2 instruments (ISO 11092, MVTR, air permeability, KES/FAST, MMT) plus the protective fabric tests of §3.	Fast, repeatable, comparable. Misses fit, layering, air gaps, movement and the wearer.
2 · Garment system	**Thermal manikin** (heated, sometimes sweating, full body in a climate chamber) measures whole-ensemble insulation (clo) and evaporative resistance; **flame manikin** measures predicted burn injury for protective ensembles.	Captures layering, fit and closures as a system. Still no perception, movement realism limited.
3 · Wearer	Controlled wearer trials: subjects work at set rates in a climate chamber (or field), reporting on subjective scales (ISO 10551) while core/skin temperature, heart rate and sweat loss are logged.	The only level that measures comfort itself and real heat strain. Slow, variable, ethically bounded (trials stop at safe core temperatures).

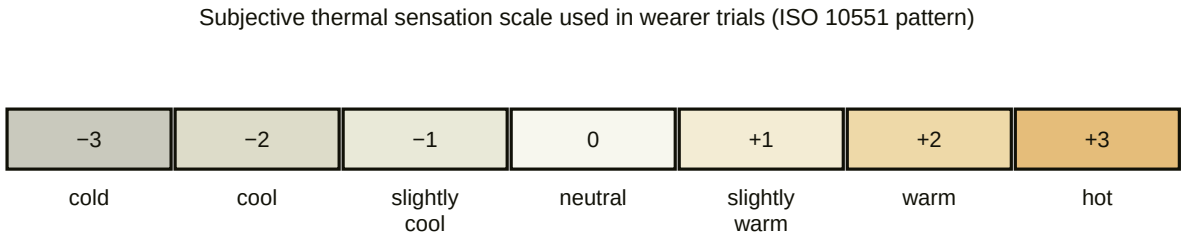


Figure 1 — The 7-point thermal sensation scale. Wearer-trial design is the discipline of making such votes comparable: same work rate, climate, exposure time and subject panel conditions.

2. Comfort evaluation in practice

A defensible comfort evaluation combines: (i) the fabric parameter set of Topic 2, with pass marks written per product (e.g. next-to-skin knit: Ret ≤ 13 class region on the garment system, high one-way transport index, air permeability within the wind-protection brief); (ii) manikin insulation for the ensemble in tog/clo against the climate/activity case computed from Topic 5's heat balance; (iii) a wearer trial whenever the product is new, protective, or premium-priced — because Levels 1–2 cannot vote. Results are reported as a **profile**, never collapsed into one "comfort score": a rain jacket and a marathon shirt can share an Ret yet be opposite products.

3. Protective performance: the standard test battery

PROPERTY	METHOD (STANDARD)	WHAT IS REPORTED
Limited flame spread	ISO 15025 (in EN ISO 11612 code A)	Afterflame/afterglow time, hole formation, debris — pass/fail criteria per class A1/A2.
Convective heat	ISO 9151 (code B)	Heat transfer index HTI24 — time for a 24 °C rise under a flame; graded B1–B5.
Radiant heat	ISO 6942 (code C)	RHTI24 at set heat flux; graded C1–C4.
Contact heat	ISO 12127 (code F)	Threshold time at contact temperatures (e.g. 250 °C); graded F1–F3.
Molten splash	ISO 9185 (codes D/E)	Mass of molten aluminium (D) or iron (E) splashed without skin damage in the film test; graded D1–D3, E1–E3.

Ensemble burn prediction	Instrumented flame manikin, ISO 13506	Predicted % body burn (2nd/3rd degree) after a standard flash exposure — a garment-system result, not a fabric one.
Chemical permeation	EN ISO 6529	Breakthrough time and permeation rate per chemical — the chemical suit's real rating.
Mechanical risks (gloves)	EN 388: Martindale abrasion; coupe cut test; tear; puncture; TDM cut (ISO 13997)	Levels: abrasion 1–4 (≥100/500/2,000/8,000 cycles); coupe cut index levels 1–5 (≥1.2/2.5/5.0/10.0/20.0); TDM reported as letters A–F by force in newtons.
Rain protection	EN 343	Two classes on the label: resistance to water penetration (hydrostatic pressure before/after pretreatments) and water-vapour resistance Ret — waterproofness and breathability graded separately.
Cold protection	EN 342 (manikin)	Resultant effective insulation I_{cler} of the ensemble, with air permeability class — matched to temperature/activity tables in the standard.
Electrostatic	EN 1149-1/-3 (surface resistance / charge decay)	Surface resistivity or half-decay time limits for dissipative protective clothing.

4. Worked reading of a test report

READING TWO GLOVE REPORTS (EN 388) LIKE A BUYER

Glove A: abrasion level 3 (≥ 2,000 cycles), coupe cut level 2 (index ≥ 2.5), tear 3, puncture 2. Glove B: abrasion level 4 (≥ 8,000 cycles), coupe cut level 5 (index ≥ 20.0), tear 4, puncture 4 — *but* B's high coupe score was measured on a blade that the fabric blunts during the test (a known coupe-test artefact with HPPE/steel yarns), so the standard requires the TDM result to govern: B's TDM letter is C (≥ 10 N). Correct specification practice: for high-cut-risk work, write the **TDM letter** into the order, treat coupe levels on blunting yarns as indicative only, and compare abrasion separately — A vs B differ fourfold in wear life (2,000 vs 8,000 cycles) whatever the cut story says.

5. Rules of honest evaluation

- **Test the article, not only the fabric:** seams, closures and ageing (after the standard's pre-treatments — washing, abrasion, flexing) are inside every PPE standard for a reason.

- **Report the level, not the adjective:** "flame resistant" is marketing; "EN ISO 11612 A1 B1 C1" is a specification.
- **Comfort is part of protective performance:** a garment so hot or stiff that workers remove it protects nobody; evaluation therefore always pairs the protective battery with the comfort ladder of §1.
- **Retest on change:** fibre blend, finish, supplier or construction changes void the old report — certification follows the tested article.

Glossary

TERM	MEANING
Thermal manikin	Heated (often sweating) full-body form measuring ensemble insulation and evaporative resistance in a climate chamber.
HTI / RHTI	Heat transfer indices: time for a defined temperature rise behind the fabric under convective (flame) or radiant exposure.
Coupe test / TDM	Two cut-resistance methods: rotating-blade index test, and the straight-blade force test (ISO 13997) used when blades are blunted.
ISO 10551	Standard for subjective judgement scales of thermal environments used in wearer trials.
Pre-treatment	Standardised washing/wear simulation applied before testing so results describe the used article, not the showroom one.

Key points and summary

- Evaluation climbs fabric → manikin → wearer; each level catches failures the level below cannot see.
- Protective claims resolve into named methods and graded levels — flame codes A–F, EN 388 levels, breakthrough times, EN 343's paired classes.
- Method artefacts matter: the coupe cut test overstates blunting yarns; TDM governs there.
- A protective evaluation that ignores comfort and heat strain is incomplete — unworn PPE has zero performance.

Practice

Answer in your own words; brief model answers follow.

1. A fabric passes ISO 11092 with Ret = 5.5, yet wearers call the jacket clammy. Give two system-level reasons the fabric result did not predict the garment.
2. State what HTI24 measures and why a time (not a temperature) is the reported quantity.
3. Two gloves both claim "cut level 5". What further information decides between them for handling sharp sheet metal?
4. Why do PPE standards insist on testing after pre-treatments such as washing?
5. Design the minimum evaluation plan (levels and tests) for a new firefighter turnout ensemble.

Answer key

1. Examples: taped seams and closures add impermeable area the fabric test never sees; fit/layering traps saturated air and the air gap behaviour differs in motion (pumping can help or hurt); the jacket's lining may have a high Ret that dominates the system. The manikin level exists to catch exactly these.
2. HTI24 is the time for the sensor behind the fabric to rise 24 °C under the standard flame — time is what protects: it predicts how long the wearer has to escape before burn injury, which a single temperature reading cannot.
3. The TDM (ISO 13997) result in newtons / letter grade — coupe levels are unreliable for yarns that blunt the rotating blade. Also compare abrasion, tear and puncture levels for the actual task, and grip/dexterity for wearability.
4. Protection must survive service: FR finishes wash out, membranes crack with flexing, coatings abrade. Testing the pre-treated article certifies the garment as worn in month six, not as sold on day one.
5. Fabric level: ISO 15025 flame spread, ISO 9151/6942 heat transfer, ISO 11092 Ret of the layer stack. System level: ISO 13506 flame-manikin burn prediction on the full ensemble, thermal-manikin insulation. Wearer level: controlled heat-stress trial with physiological monitoring and ISO 10551 votes, within ethical stop limits — plus the standard's design, sizing and after-care checks.
